Emergence, Onset Timing, and Predictors of Probable Insomnia in Low-Symptom Pregnant Individuals Across Pregnancy and Postpartum

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BACKGROUND

Perinatal insomnia is associated with adverse mental health and birth outcomes. However, **little is known about who is most at risk.**

We examined the **emergence** and **course of probable insomnia** from late pregnancy to one year postpartum among those with low symptoms (< 8 Insomnia Severity Index [ISI]) in late pregnancy.

SAMPLE & METHOD

321 **pregnant women**
Age: 33.6 ± 3.8 years
Low insomnia symptoms at late pregnancy

Outcome: Probable insomnia (categorised as ISI ≥ 10)

PREGNANCY — POSTPARTUM

26-32 weeks · 31-37 weeks · 3 months · 6 months · 12 months

Predictors:

- Maternal Age
- Existing Children (Parity)
- Education (Above bachelor vs less than bachelor)
- Dysfunctional Beliefs About Sleep
- Sleep Reactivity (Ford Insomnia Response to Stress Test)

Analysis: Multinomial regressions, using the Never Insomnia group as the reference, examined predictors of trajectory.

TRAJECTORY OF PROBABLE INSOMNIA

Four trajectories were examined **(Figure 1)**:

- **Never Insomnia (n=139):** never met probable insomnia threshold
- **Insomnia Recovery (n=107):** met threshold in pregnancy and/or early postpartum but recovered by 12-months-postpartum
- **Persistent Insomnia (n=51):** met threshold in pregnancy and/or early postpartum which persisted at 12-months-postpartum
- **Late Postpartum-Onset Insomnia (n=24):** first met threshold at 12-months-postpartum

MULTINOMIAL REGRESSIONS

Compared to "Never Insomnia"	Insomnia Recovery	Persistent Insomnia	Late Postpartum-Onset
Dysfunctional Beliefs About Sleep	**1.36**	**1.73**	**1.73**
Sleep Reactivity	**1.05**	**1.06**	—
Existing Children (Parity)	—	**2.05**	—
Age & Education	Not associated with any trajectory		

Relative risk ratios (see Figure 2). Coloured cells = statistically significant; — = not significant.

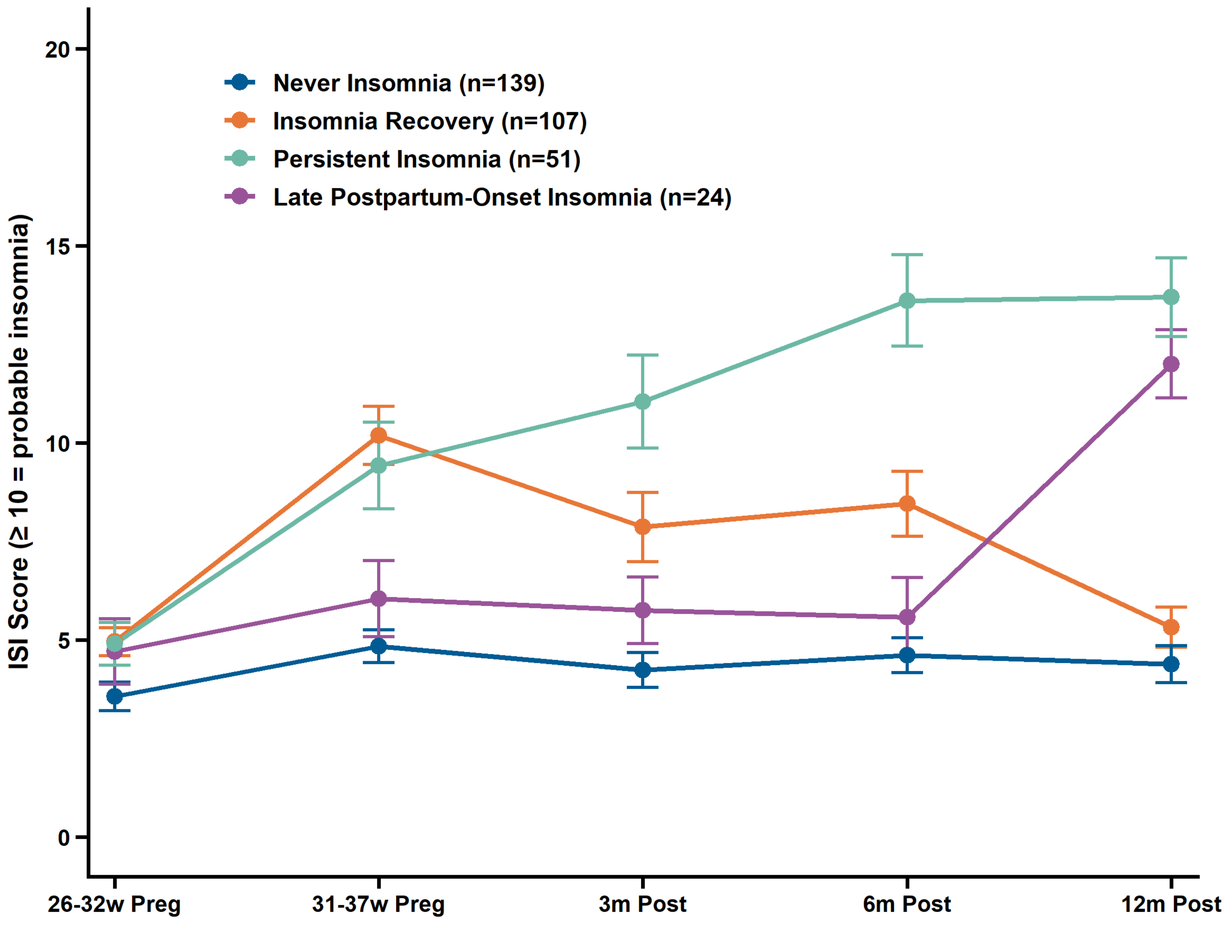


Figure 1. Trajectory of insomnia symptoms (ISI) from late pregnancy (26-32 weeks) to 12 months postpartum, by trajectory of probable insomnia.

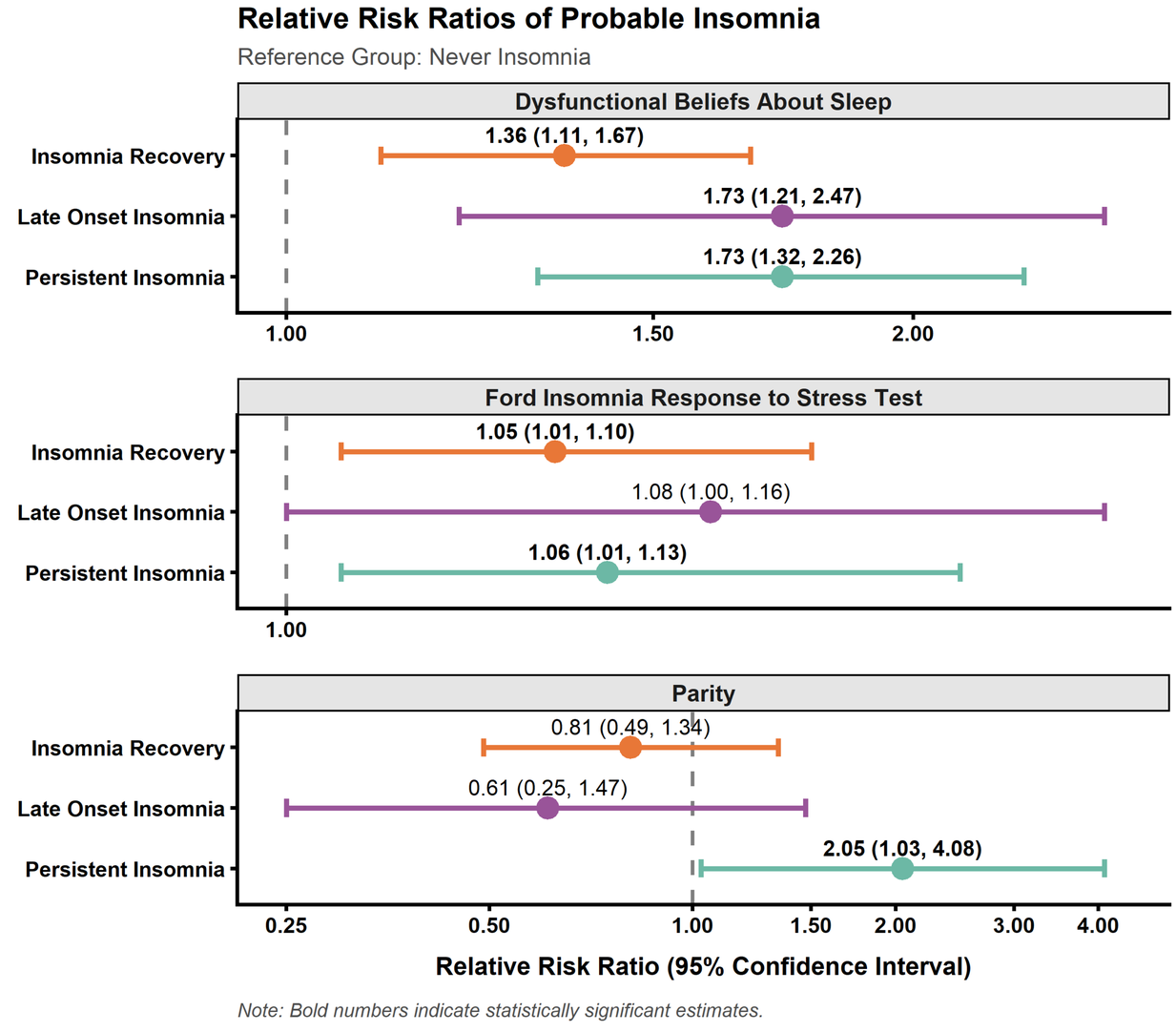


Figure 2. Forest plot of the Relative Risk Ratio & 95% Confidence Intervals for Multinomial Regressions of predictors of insomnia trajectory.

Among women with low insomnia symptoms in late pregnancy, **distinct insomnia trajectories emerge.**

Key risk factors

Dysfunctional sleep beliefs and sleep reactivity may be key risk factors for insomnia development.

Existing children

Having existing children may increase vulnerability to persistent symptoms.

Early prevention

Targeting unhelpful beliefs, and identifying individuals with heightened sleep reactivity may offer opportunities for early prevention of perinatal insomnia.

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